Series 1 – *Reminders on Basics of Algorithms*

Exercise 1

To perform the cubic root of a real number $A \geq 0$ we can use the following iterative method: We take an initial value $x1 = A/3$ then we search for successive items $x2$, $x3$, ... of the sequence defined by: $x_{n+1} = \frac{2x_n + \frac{A}{x_n^2}}{3}$

After each iteration, we compare x_{n+1} to x_n and we stop the process when $|x_{n+1} - x_n| < 10^{-6}$.
In this case, x_n represents the cubic root of A. Write an algorithm to provide the cubic root of a real number A using the described method below.

Exercise 2

Let A be a one-dimension array containing N integers (where N≤100). Write an algorithm to delete all the null values from the array A.

Exercise 3

Consider S a string. Write an algorithm to clean the string S from all non-alphabetical characters but keep the spaces (blank character).
Example: **S= "*/ The cours(@e o+f Al?!gor><ithm&s is °$ interesting "**

Cleaned S = "The course of Algorithms is interesting "

Exercise 4

Write an algorithm that calculates: $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!}$
using two functions: one function to calculate the factorial and another to calculate the power.

Exercise 5

Consider the following algorithm:

```
Algorithm exercise_5 ;
Var A, B: real;
Procedure Objective (X : real, Y : real) ;
Begin                                              R1
        X ← X*Y ;   Y ← X/Y ;   X ← X/Y ;
End;
Function Capture (X: real, Y: real): real ;
Var Z: real;
Begin
        If X > Y Then  Z ← X ;                     R2
        Else Z ← Y ; EndIf;
        Return Z ;
End ;
Begin
    Write ("Give the values of A and B") ;   Read (A, B) ;
        If Capture (A, B) = B Then                 R3
           Objective (A, B) ;
        EndIf ;
        Write ("A= ", A, " and B= ", B) ;
End.
```

Questions: (we read 5 and 11 for A and B successively)

1- Determine the formal and real parameters used in this algorithm.
2- For the function, determine the local variables and the global variables.
3- Consider regions R1, R2 and R3: what is the scope of each of the variables: A, B and Z?
4- What does the Objective procedure do?
5- What do A and B contain after calling the Objective procedure?
6- What is missing so that the procedure returns the correct values after its call?
7- What is the role of the Capture function?
8- What does the algorithm do?

Exercise 6

Write an algorithm that creates a dynamic array of size N entered by the user, then using a procedure, fill this array and using a function, calculate and return the average of the elements of the array (use a pointer to traverse the array).